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Seeing, feeling, and buying: a mixed-methods investigation of emotional wellbeing in augmented reality e-commerce

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Abstract

As e-commerce platforms increasingly adopt augmented reality (AR), understanding how immersive technologies influence consumer wellbeing has become essential. Drawing on the stimulus-organism-response (S-O-R) model and cognitive appraisal theory, this article develops and empirically tests a framework that examines how AR features shape affective satisfaction, buyer empowerment, emotional wellbeing, and purchase intention. A mixed-methods design was employed, involving qualitative interviews and a cross-national survey of AR users. Findings indicate that personalized interactivity and usability realism strengthen immersive experiences and perceived value, which subsequently drive emotional outcomes and behavioral intentions. Moderating effects of perceived artificial intelligence (AI) intrusiveness and AR-related anxiety suggest that psychological discomfort can diminish satisfaction's impact on purchase intention. Overall, the study integrates cognitive and emotional psychological pathways to extend digital marketing theory and offers actionable insights for designing e-commerce experiences that support consumer wellbeing and ethical engagement, understood as the protection of consumer autonomy, privacy, and psychological comfort during immersive shopping, in e-commerce.

Keywords Augmented reality · Consumer wellbeing · Stimulus-organism-response · Cognitive appraisal · E-commerce · Purchase intention

1 Introduction

Augmented reality (AR) has emerged as a powerful driver of transformation in online retail, as continuous digital advancements reshape how shoppers interact with products. More broadly, e-commerce has evolved from a transaction-focused channel into a comprehensive digital ecosystem that emphasizes convenience, personalization,

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and immersive customer experiences [1–4]. This technology allows consumers to visualize items in their real-world surroundings, explore product features from different perspectives, and personalize products in highly engaging ways. E-commerce sales surpassed $5.8 trillion in 2023 and are expected to exceed $8 trillion by 2027 [5]. AR holds significant promise for enhancing online purchasing experiences by facilitating interactive product displays and simulated trials in an environment that transcends conventional digital offerings [6]. Earlier work has emphasized how AR can boost engagement, enjoyment, and purchase intent [7].

Many studies have focused on practicality gains such as convenience and satisfaction, yet potential emotional drawbacks remain underexplored [6, 8, 9]. Immersive interactions can generate anxiety, discomfort, and impressions of reduced control, particularly when AI-enabled personalization accompanies these technologies. Although some attention has been devoted to these concerns, deeper understanding of AR's cognitive and emotional psychological impacts is still lacking. Failing to account for these issues may undermine consumer wellbeing and ethical engagement.

This article employs a mixed-methods exploratory design to address these gaps. The initial phase consisted of in-depth interviews with 20 online shoppers, capturing their cognitive and emotional psychological reflections on AR. Insights from these interviews informed the development of a conceptual model grounded in the stimulus-organism-response (S-O-R) model [10] and cognitive appraisal theory [11]. This model incorporates personalization, usability, immersiveness, intrusiveness, and anxiety as key AR elements that influence perceived value, affective satisfaction, buyer empowerment, emotional wellbeing, and purchase intention. Additional perspectives on immersive technologies, psychological responses, and ethics in digital contexts [8, 12–14] provided further conceptual grounding. The second phase involved surveying 477 international AR users to statistically evaluate the proposed model.

This article advances understanding by integrating theoretical insights with first-hand accounts of AR usage. Findings shed light on how specific AR attributes can foster or hinder emotional wellbeing, pointing to the significance of responsible design and ethical engagement in digital commerce. Ethical engagement here refers to designing AR that respects consumer autonomy, privacy, and psychological comfort rather than exploiting attention or personal data, a concern the study operationalizes through perceived artificial intelligence (AI) intrusiveness, AR-induced anxiety, and buyer empowerment. Embracing these considerations can help organizations craft immersive experiences that appeal to consumers' aspirations for convenience and enjoyment while acknowledging wider psychological and social implications. This article thereby enriches discussions on digital ethics, transformative service, and potential over-personalization in online retail contexts, a connection that is concrete rather than rhetorical, whereby the moderating role of perceived intrusiveness speaks to digital ethics and over-personalization, whereas the modeling of emotional wellbeing as a central outcome speaks to transformative service research, which treats consumer wellbeing as a legitimate end of service design [15].

2 Literature review

2.1 Stimulus-organism-response (S-O-R) model

The S-O-R model has served as an influential approach for examining how technology and environmental elements influence consumer behavior through emotional and cognitive pathways. Grounded in environmental psychology [10], this framework posits that behavioral responses (R) stem from internal states (O), which are shaped by external stimuli (S). Researchers have adapted S-O-R to explore how design factors, interface features, and technological capabilities can sway users' feelings and decisions in digital and service contexts. Hoyer et al. [16] contend that immersive technologies, such as AR interfaces, can trigger a spectrum of psychological responses, from feeling overwhelmed to experiencing deep enjoyment. In e-commerce contexts, these stimuli frequently involve interactivity, personalization, and realism, each significantly impacting perceived value and enjoyment [9]. The organism phase, typically encompassing emotional and cognitive appraisals, often includes affective satisfaction and experiential value [7]. These internal states guide responses such as purchase intention, loyalty, empowerment, and wellbeing, highlighting S-O-R's efficacy for understanding how immersive design can influence behavior.

2.2 Cognitive appraisal theory

Although S-O-R delineates a general route from environment to behavior, it does not specify the cognitive mechanisms that mediate emotional and behavioral outcomes. Cognitive appraisal theory [11] addresses this gap by proposing that emotional responses do not emerge from stimuli alone; they stem from an individual's evaluation of those stimuli. This process involves two appraisals: a primary appraisal, which judges whether a stimulus is relevant or threatening to an individual's goals, and a secondary appraisal, which assesses whether the individual can cope with or benefit from that stimulus. In digital settings, users' risk perceptions and desired outcomes shape these appraisals. Yin et al. [17] observe that AI-based personalization can be seen as invasive if users appraise the experience as intrusive, sparking discomfort or constrained choice. In contrast, AR features that are appraised favorably may induce competence and satisfaction [18].

A combined perspective of S-O-R and cognitive appraisal theory yields deeper insights into the ways AR influences consumer wellbeing, either by enhancing or hindering engagement and emotional welfare. The S-O-R framework highlights how AR functions as an external stimulus driving users' reactions while cognitive appraisal theory clarifies the internal evaluations that determine whether the stimulus is beneficial or stressful, enabling a matching and parsimonious account of the studied phenomenon [19]. Moderators such as technological invasiveness further shape users' appraisals, reflecting calls for more emotion-aware technology design [8]. The next section outlines the qualitative study that investigates AR's psychological effects on consumer wellbeing in the context of online shopping.

3 Study 1 - qualitative exploratory study

3.1 Approach

This exploratory phase clarifies how AR influences consumers' cognitive and emotional psychological responses in online shopping contexts. A qualitative approach was adopted to investigate immersive commerce experiences, involving 20 online shoppers from the United States, Germany, South Korea, and India. All participants had used AR features within the previous six months to shop for items such as apparel, cosmetics, electronics, or furniture. Interviews were conducted in English or a local language, then transcribed and translated to maintain consistency. The selection of New York, Berlin, Seoul, and Hyderabad was based on their strong e-commerce infrastructures and substantial AR adoption, enabling cross-cultural perspectives (Table 1). More specifically, New York, Berlin, and Seoul sit in advanced, high-income digital economies with well-established e-commerce [20] and relatively mature AR adoption [21], whereas Hyderabad was included deliberately as a contrasting case from a high-growth emerging market. India ranks among the fastest-expanding e-commerce markets in the world [22] and the country's AR market is projected to expand at one of the steepest rates globally on the strength of retail and e-commerce applications [23]. Hyderabad, in turn, ranks as India's second-largest software-exporting city and one of its leading technology hubs [24]. Therefore, pairing Hyderabad with the three mature markets allows the qualitative phase to capture experiences across both established and rapidly developing AR contexts, which, in turn, widens the cross-cultural reach of the study and prevents the emergent themes from reflecting only affluent Global North usage [25].

3.2 Methods

The interview protocol followed [26] and is reproduced in full in the Appendix, featuring both structured and open-ended queries, allowing participants to share cognitive and emotional psychological reflections on AR-enhanced e-commerce. Minor cultural adjustments ensured comprehension without altering core content. Interviews were conducted in person or virtually, depending on participants' preferences and researchers' availability. An inductive thematic analysis was applied, following open, axial, and selective coding phases. Intercoder reliability exceeded 0.80, confirming analytic consistency. A cross-cultural lens illuminated shared themes and culture-specific nuances in AR usage toward theoretical saturation [27]. Triangulation with reviews from major AR platforms reinforced the emergent findings.

Table 1 Overview of interviews

Location	Number of interviews	Total duration
New York	5	5.5 h
Berlin	5	6 h
Seoul	5	5.5 h
Hyderabad	5	5 h
Total	20	22 h

3.3 Results

Ten principal themes arose from the analysis, forming the basis of the proposed framework: AR personalized interactivity, usability realism, immersive experience, perceived value, affective satisfaction, perceived intrusiveness, induced anxiety, empowerment, emotional wellbeing, purchase intention (Figs. 1 and 2). AR's personalized interactive features (personalization options, user control and flexibility) and usability realism (visual realism, system usability) contributed to its immersive experience (cognitive absorption, emotional engagement) and perceived value (functional utility, perceived worthiness), which, in turn, determine users' affective satisfaction (hedonic enjoyment, positive emotional response). This satisfaction contributes to users—as buyers—empowerment (decision-making confidence, perceived autonomy), emotional wellbeing (emotional comfort, psychological positivity), and purchase intention (purchase readiness, behavioral attachment), though its impact on the latter may be influenced by perceived intrusiveness (personalization intrusiveness, privacy concerns) and induced anxiety (sensory overload, interaction anxiety) in AR e-commerce environments driven by AI. Indeed, concerns about AI intrusiveness and over-personalization in AR appeared frequently among participants in Hyderabad and New York, with descriptions such as "creepy" or "overwhelm-

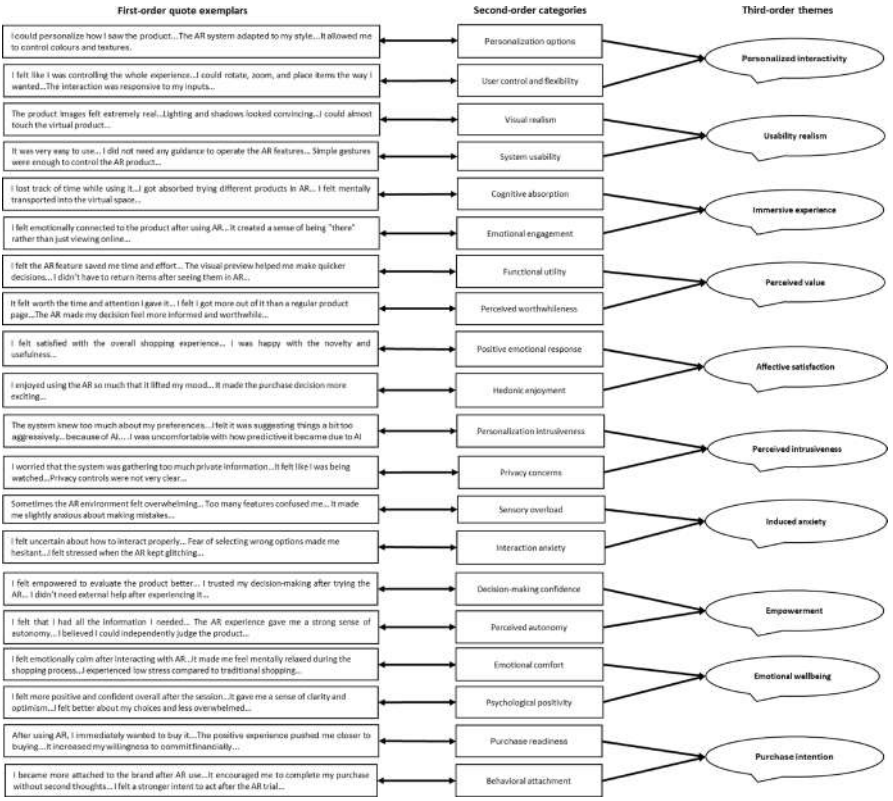


Fig. 1 Thematic development

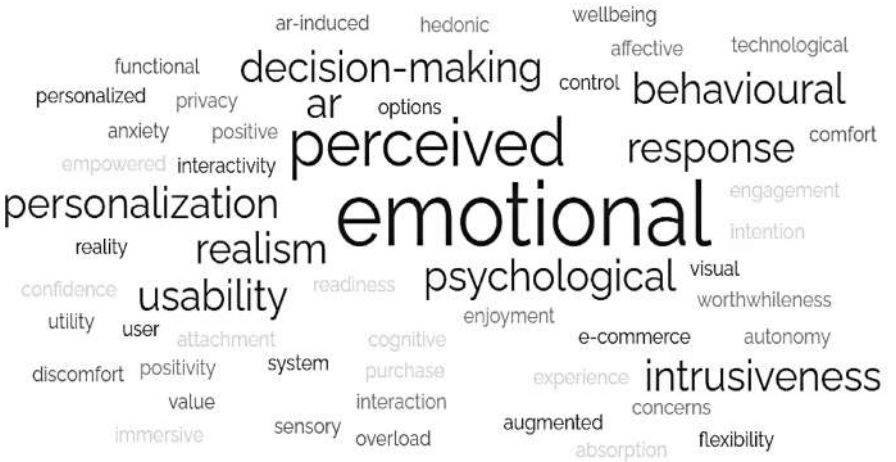


Fig. 2 World cloud

ing." Emotional discomfort and anxiety were more pronounced among shoppers less familiar with digital technologies or highly attentive to privacy while participants who sensed greater control over AR interactions reported stronger emotional satisfaction, empowerment, and readiness to purchase.

These observations cut across cultural contexts and underscored the complexity of emotional and cognitive factors in AR-driven e-commerce shopping. Guided by these findings and extant research [14, 28, 29], a conceptual framework was developed, drawing on the S-O-R model and cognitive appraisal theory. Within that framework, AR features serve as stimuli that influence internal cognitive appraisals and subsequent emotional states, leading to behavioral outcomes. Perceived AI intrusiveness and AR-induced anxiety were included as moderating influences, based on both the interviews and existing scholarship. This qualitative phase thus clarifies how consumers interpret and emotionally respond to AR, informing a theory-driven framework that integrates real-world viewpoints. The next follow-up study extends these findings through a large-scale survey.

4 Study 2 – quantitative confirmatory study

4.1 Hypothesis development

4.1.1 AR personalized interactivity, immersive experience, perceived value

AR applications that offer personalized interactivity tend to enhance a user's control and engagement, allowing for meaningful customization of the experience. This heightened personalization fosters involvement and presence, which can generate emotional resonance and relevance [7]. Interactivity promotes user engagement, while personalization increases relevance and elicits strong emotional responses

[30]. Co-creation shifts users from passive recipients to active participants, prompting them to commit more cognitive and affective resources to the AR environment. Personalization embedded in AR also boosts perceived value, especially when users notice a close alignment between the technology's functionality and their individual needs and preferences [31]. Tailored interactions can amplify both enjoyment and perceived utility, reinforcing the cognitive and emotional evaluations that drive positive reactions. Such responses are influential in online retail, where perceived value guides satisfaction and purchase decision processes through a sense of personal relevance and agency. Based on these arguments, the following hypotheses are proposed:

H_{1a} *AR personalized interactivity positively influences immersive experience.*

H_{1b} *AR personalized interactivity positively influences perceived value.*

4.1.2 AR usability realism, immersive experience, and perceived value

Usability, defined by intuitive interfaces and manageable cognitive demands, enables users to remain engaged and present during AR sessions [32]. Realism, operationalized through lifelike visuals, credible spatial representations, and stable sensory feedback, promotes an environment where users can suspend disbelief and experience an impression of presence [33]. In combination, strong usability and realism have been linked to deeper immersion and greater satisfaction. Weak interface designs or unrealistic simulations can negate these benefits, reducing users' immersion and dampening their overall evaluations. Clear alignment of usability and realism fosters trust, functional benefits, and emotional fulfillment, which heightens perceived value in technology-mediated commerce [18, 34, 35]. Shoppers who navigate a fluid and believable AR environment can interpret product information more effectively, strengthening the user's perception of both usefulness and enjoyment. Based on these considerations, the following hypotheses are proposed:

H_{2a} *AR usability realism positively influences immersive experience.*

H_{2b} *AR usability realism positively influences perceived value.*

4.1.3 Immersive experience and perceived value

Immersive experiences stand as key precursors to users' perceived value in e-commerce. Heightened involvement and emotional connection increase perceptions of functional and hedonic value [36]. An immersive AR environment facilitates exploration, trial, and interactive product assessment, improving the user's ability to make informed and rewarding decisions. Realism, interactivity, and multisensory engagement often strengthen perceptions of utility, pleasure, and relevancy, reinforcing perceived value [37]. These qualities become especially vital in e-commerce, where users rely on immersive cues to substitute for physical product interactions. Reduced uncertainty and heightened engagement arising from immersion are likely to augment perceived value. Consequently:

H_3 *Immersive experience positively influences perceived value.*

4.1.4 Perceived value and affective satisfaction

Perceived value exerts substantial influence on users' emotional reactions to AR applications, given its functional and emotional components. Individuals who perceive AR technology as worthwhile, relevant, and rewarding typically report stronger affective satisfaction [38]. Positive perceptions of utility and pleasure reduce anxiety about time and effort spent, bolstering comfort and contentment [18, 39]. Online shoppers often feel greater certainty when they believe the technology aligns with personal preferences and objectives. This elevated sense of relevance promotes an emotionally gratifying experience. Hence:

H_4 *Perceived value positively influences affective satisfaction.*

4.1.5 Affective satisfaction, buyer empowerment, emotional wellbeing, and purchase intention

Affective satisfaction represents a critical psychological process in augmented environments that offer engagement and enjoyment [40]. Users who experience excitement or pleasure through AR may feel deeper emotional fulfillment, which encourages positive psychological outcomes, including emotional wellbeing [41]. Positive emotions often strengthen feelings of control, optimism, and confidence, contributing to a sense of empowerment. Greater empowerment heightens the user's agency, which may translate into stronger readiness to buy. Studies have linked enjoyable interactions in AR to emotional bonding, enhanced trust, and favorable purchase intentions [18]. Immersive product displays, tailored content, or interactive features can stimulate a psychological readiness to engage in commercial transactions [35]. Affective satisfaction in turn reinforces a sense of competence in decision-making, promoting buyer empowerment and a sense of personal wellbeing. In line with these observations:

H_5 *Affective satisfaction positively influences buyer empowerment.*

H_6 *Affective satisfaction positively influences emotional wellbeing.*

H_7 *Affective satisfaction positively influences purchase intention.*

4.1.6 Buyer empowerment, emotional wellbeing, and purchase intention

Buyer empowerment contributes to emotional wellbeing by enhancing perceived control, capability, and ownership within technology-based interactions. Users who feel empowered report less uncertainty and stress, particularly in immersive AR settings [35]. Greater control over product information and decision processes is associated with higher psychological satisfaction and reduced anxiety [42]. The emotional wellbeing that emerges from empowerment often leads to stronger purchase inten-

tions, as it encourages an optimistic and engaged mindset when evaluating potential purchases [43]. Favorable emotional states drive higher levels of trust and positive affect, fostering readiness to transact [44]. Therefore:

H_8 *Buyer empowerment positively influences emotional wellbeing.*

H_9 *Emotional wellbeing positively influences purchase intention.*

4.1.7 Moderating roles of perceived AI intrusiveness and AR-induced anxiety

Personalized AR is frequently powered by AI-driven recommendation systems, which can provoke perceptions of excessive control or data misuse [8]. Users who experience AI personalization as intrusive or manipulative may downgrade their willingness to purchase, even when they feel satisfied at an emotional level. Anxiety further complicates this effect by decreasing the impact of positive emotions on subsequent behaviors [45]. Shoppers who feel overwhelmed or wary of losing control within AR environments may be hesitant to convert favorable feelings into active buying decisions. Negative experiences arising from intrusiveness and anxiety can reduce the relationship between affective satisfaction and purchase intention. Thus:

H_{10a} *Perceived AI intrusiveness negatively moderates the relationship between affective satisfaction and purchase intention.*

H_{10b} *AR-induced anxiety negatively moderates the relationship between affective satisfaction and purchase intention.*

The research framework that illustrates the S-O-R model as the theoretical framework and the conceptual relationships as the conceptual framework is presented in Fig. 3.

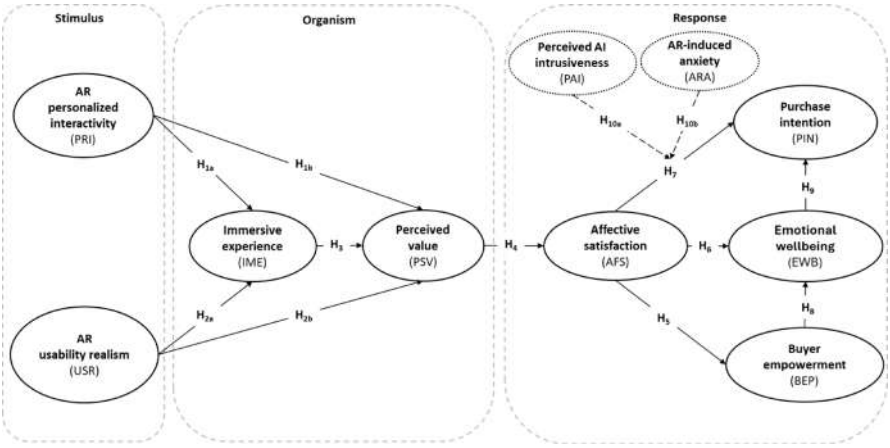


Fig. 3 Research framework

4.2 Methods

This study examines the ways in which AR features in e-commerce can shape consumer behavior. Data were gathered from diverse cultural contexts, reflecting the global reach of e-commerce and varying AR adoption patterns. A structured online survey (Table 3) was conducted using purposive sampling to target users who had experienced AR in their online shopping within the last six months. Two screening questions ensured only qualified participants proceeded: (a) whether they had used AR features while shopping online, and (b) whether those features offered interactivity or personalization. Respondents who responded "Yes" to both questions were included. The survey link was shared via academic and professional networks, digital marketing panels, online shopping forums, and social media ads to reach shoppers familiar with AR. Measures such as single-entry authentication and required fields reduced incomplete responses, yielding 477 valid responses.

Armstrong and Overton's [49] procedures were employed to assess possible non-response bias by comparing early and late respondents on key demographic and attitudinal variables. No significant differences arose, suggesting minimal non-response concerns. Confirmatory factor analysis (CFA) and structural equation modeling (SEM) were then performed using AMOS *v.*24 to evaluate measurement reliability, validity, and the hypothesized structural model.

Participants consisted of 225 females (47.17%), 248 males (51.99%) while 4 respondents (0.84%) who identified as either non-binary or chose not to disclose (Table 2). The largest age cohort (38.16%) fell between 25 and 34 years, followed by 35 to 44 years (25.79%). Roughly half of the sample resided in North America or Europe, while the remaining half lived in South or East Asia, reflecting balanced representation across regions. Monthly income levels ranged from under USD 1,000 (29.56%) to over USD 2,500 (28.93%), illustrating economic diversity. Fashion and apparel (31.66%) represented the most common AR-shopping category, followed by electronics (25.57%), furniture (22.85%), and cosmetics (19.92%). Respondents reported experiences with a variety of AR features, from basic visual overlays (21.38%) to interactive try-ons (45.91%) and fully personalized AR interfaces (32.71%). This breadth of demographic and shopping backgrounds facilitated a well-rounded exploration of how AR shapes emotional and behavioral outcomes in global e-commerce.

4.3 Results

4.3.1 Preliminary checks

Preliminary checks were conducted in line with Lim [50]. The dataset was screened to verify the assumptions of normality, with all skewness and kurtosis values lying within ± 2. Multicollinearity was examined through variance inflation factor (VIF) and tolerance values, wherein all VIFs were below 5, indicating no multicollinearity concerns. Descriptive statistics and inter-construct correlations were generated to explore associations among the study variables, forming a solid basis for further confirmatory and structural analyses. Common method bias (CMB) was tested using

Table 2 Profile of survey respondents

Socio-demographic	Category	*n* (477)	% (100)
Gender	Female	225	47.17%
	Male	248	51.99%
	Prefer not to say/Other	4	0.84%
Age group	Less than 25 years	66	13.84%
	25–34 years	182	38.16%
	35–44 years	123	25.79%
	45–54 years	67	14.05%
	55 years and above	39	8.17%
Region	Global North (e.g., America, Europe)	233	48.85%
	Global South (e.g., East Asia, South Asia)	244	51.15%
Monthly income	Less than USD 1,000	141	29.56%
	USD 1,000–2,500	198	41.51%
	More than USD 2,500	138	28.93%
AR shopping category	Fashion and apparel	151	31.66%
	Furniture and home decor	109	22.85%
	Electronics and gadgets	122	25.57%
	Cosmetics & personal care	95	19.92%
Type of AR experience	Basic visual overlay	102	21.38%
	Interactive try-on/ visualization	219	45.91%
	Fully personalized AR interface	156	32.71%

Harman's single-factor procedure, revealing that the first factor explained only 28.6% of total variance, which is well below the 50% threshold. This result suggests that common method bias was not a significant issue for this study.

4.3.2 Measurement model

Confirmatory factor analysis (CFA) was employed to assess the psychometric properties of the constructs and their measurement items (Table 3) in line with Lim [50]. Composite reliability (CR) ranged from 0.806 to 0.941, indicating strong *internal consistency*. Likewise, *convergent validity* was confirmed, as each construct's average variance extracted (AVE) exceeded 0.50 while *discriminant validity* was established via the Fornell-Larcker criterion: for each construct, the square root of its AVE was greater than its correlations with other constructs, and the Maximum Shared Variance (MSV) was lower than the AVE (Table 4).

The overall measurement model achieved an excellent fit (Table 5). The ratio of chi-square to degrees of freedom ($\chi 2/df$) was 1.799, which is under the maximum threshold of 3. The comparative fit index (CFI) was 0.968, exceeding the minimum threshold of 0.90. the root mean square error of approximation (RMSEA) was 0.042 and the standardized root mean square residual (SRMR) was 0.044, both well below the maximum threshold of 0.08. A PClose value of 0.997 further affirms the model's strong fit.

Table 3 Measurement items

Construct	Item statement	Item code	CFA loading	SEM loading	Source
AR personalized interactivity (PRI)	I can customize the AR experience according to my preferences.	PRI1	0.698	0.69	Alimamy and Gnoth, [31]
	The AR features respond effectively to my inputs.	PRI2	0.825	0.832	
	I feel I have control over how the AR experience unfolds.	PRI3	0.807	0.812	
	The AR system adapts to my needs and choices.	PRI4	0.73	0.722	
AR usability realism (USR)	It is easy to navigate the AR experience.	USR1	0.718	0.709	Daassi and Debbabi, [32]
	The AR visuals appear lifelike and realistic.	USR2	0.729	0.747	
	The AR features work smoothly without interruptions.	USR3	0.749	0.756	
	Interacting with the AR felt natural and intuitive.	USR4	0.655	0.636	
Immersive experience (IME)	I felt completely absorbed in the AR environment.	IME1	0.776	0.775	Vo et al. [45]
	I lost track of time while engaging with the AR experience.	IME2	0.966	0.966	
	The AR experience made me feel like I was "inside" the environment.	IME3	0.997	0.997	
Perceived value (PSV)	Using the AR experience is worth my time and effort.	PSV1	0.755	0.755	Trivedi et al. [35]
	The AR features add significant value to my shopping experience.	PSV2	0.742	0.747	
	I find the AR experience to be beneficial and useful.	PSV3	0.791	0.778	
Affective satisfaction (AFS)	I am emotionally satisfied with the AR shopping experience.	AFS1	0.791	0.784	Iranmanesh et al. [34]
	I enjoyed using the AR features during shopping.	AFS2	0.76	0.762	
	I felt pleased with the overall AR experience.	AFS3	0.815	0.816	
	I felt happy engaging with the AR environment.	AFS4	0.81	0.813	
Perceived AI intrusiveness (PAI)	I felt that the AR experience invaded my privacy.	PAI1	N/A	N/A	Ameen et al. [46], Puntoni et al. [47]
	I felt the AR personalization features were too intrusive.	PAI2	N/A	N/A	
	I felt uncomfortable with how much the AR system knew about me.	PAI3	N/A	N/A	
	I felt the AR recommendations were manipulative.	PAI4	N/A	N/A	

Table 3 (continued)

Construct	Item statement	Item code	CFA loading	SEM loading	Source
AR-induced anxiety (ARA)	I felt anxious when using the AR virtual try-on feature.	ARA1	N/A	N/A	Chang et al. [48]
	The AR experience made me feel uneasy while shopping.	ARA2	N/A	N/A	
	I felt nervous interacting with the AR environment.	ARA3	N/A	N/A	
	I felt stressed when using the AR features.	ARA4	N/A	N/A	
Buyer empowerment (BEP)	I feel more confident making purchase decisions after using the AR.	BEP1	0.875	0.875	Abou-Shouk et al. [37]
	The AR experience made me feel more in control of my shopping.	BEP2	0.86	0.86	
	I felt empowered while using the AR features.	BEP3	0.894	0.895	
	The AR technology made me feel like a capable shopper.	BEP4	0.813	0.812	
Emotional wellbeing (EWB)	The AR experience made me feel positive and uplifted.	EWB1	0.942	0.939	Park et al. [41]
	I felt emotionally refreshed after using the AR features.	EWB2	0.898	0.902	
	The AR experience contributed to my overall emotional wellbeing.	EWB3	0.559	0.554	
Purchase intention (PIN)	I am likely to purchase products after using the AR experience.	PIN1	0.816	0.811	Ngo et al. [9]
	I intend to buy the products I interacted with through AR.	PIN2	0.815	0.815	
	The AR experience increased my willingness to purchase.	PIN3	0.838	0.843	
	I would consider purchasing based on my AR experience.	PIN4	0.848	0.848	

N/A = Not applicable. PAI and ARA were not included in the measurement and structural model because they were only used to examine moderation (interaction) effects (Figs. 5 and 6), not main (direct) effects

4.3.3 Structural model

The structural model was tested using SEM (Fig. 4). Structural model fit indices proved acceptable ($\chi^2/df = 2.399 < 3$, CFI = 0.939 > 0.90, RMSEA = 0.054 < 0.08), meeting the thresholds. The results supported the hypothesized relationships (Table 6). More specifically, AR personalized interactivity (H_{1a}: $\beta = 0.275$, $p < 0.001$) and usability realism (H_{2a}: $\beta = 0.189$, $p < 0.05$) exert a significant positive influence on immersive experience. Likewise, AR personalized interactivity (H_{1b}: $\beta = 0.194$, $p < 0.001$) and usability realism (H_{2b}: $\beta = 0.119$, $p < 0.10$) also exert a significant positive influence on perceived value. In turn, immersive experience significantly predict perceived value (H_3: $\beta = 0.122$, $p < 0.001$), which positively influences affective satisfaction (H_4: $\beta = 0.163$, $p < 0.01$). While affective satisfaction positively influences buyer empowerment (H_5: $\beta = 0.631$, $p < 0.001$), emotional wellbeing (H_6: $\beta = 0.169$, $p < 0.10$), and purchase intention (H_7: $\beta = 0.160$, $p < 0.05$), with buyer empowerment positively influencing emotional wellbeing (H_8: $\beta = 0.284$, $p < 0.001$) and emotional wellbeing positively influencing purchase intention (H_9: $\beta = 0.095$, $p < 0.10$), the influ-

Table 4 Measurement model statistics

Construct	Internal consistency		Convergent validity	Discriminant validity	PRI	USR	IME	PSV	AFS	BEP	EWB	PIN
	CR	MaxR(H)	AVE	MSV								
PRI	0.85	0.859	0.59	0.25	**0.767**							
USR	0.81	0.809	0.51	0.13	0.317***	**0.714**						
IME	0.94	0.995	0.84	0.06	0.232***	0.174***	**0.919**					
PSV	0.81	0.809	0.58	0.09	0.259***	0.180**	0.229***	**0.76**				
AFS	0.87	0.874	0.63	0.28	0.222***	0.365***	0.106*	0.146**	**0.794**			
BEP	0.92	0.923	0.74	0.28	0.130*	0.151**	0.106*	0.120*	0.530***	**0.861**		
EWB	0.85	0.926	0.67	0.25	0.497***	0.285***	0.156**	0.101†	0.231***	0.284***	**0.818**	
PIN	0.90	0.899	0.69	0.09	0.304***	0.226***	0.252***	0.304***	0.128*	0.093†	0.126*	**0.83**

† = $p<0.10$. * = $p<0.05$. ** = $p<0.01$. *** = $p<0.001$. CR = Composite reliability. AVE = Average variance extracted. MSV = Maximum shared variance. MaxR(H) = Maximum reliability. PRI = AR personalized interactivity. USR = AR usability realism. IME = Immersive experience. PSV = Perceived value. AFS = Affective satisfaction. BEP = Buyer empowerment. EWB = Emotional wellbeing. PIN = Purchase intention. Bold diagonal values are square roots of AVE

Table 5 Measurement model fit indices

Measure	Estimate	Threshold	Interpretation
Chi-square (χ^2)	627.999	–	–
Degree of freedom (df)	349	–	–
χ^2/df	1.799	<3.00	Excellent
Comparative fit index (CFI)	0.968	>0.90	Excellent
Root mean square error of approximation (RMSEA)	0.042	<0.08	Excellent
Standardized root mean square residual (SRMR)	0.044	<0.08	Excellent
PClose	0.997	>0.05	Excellent

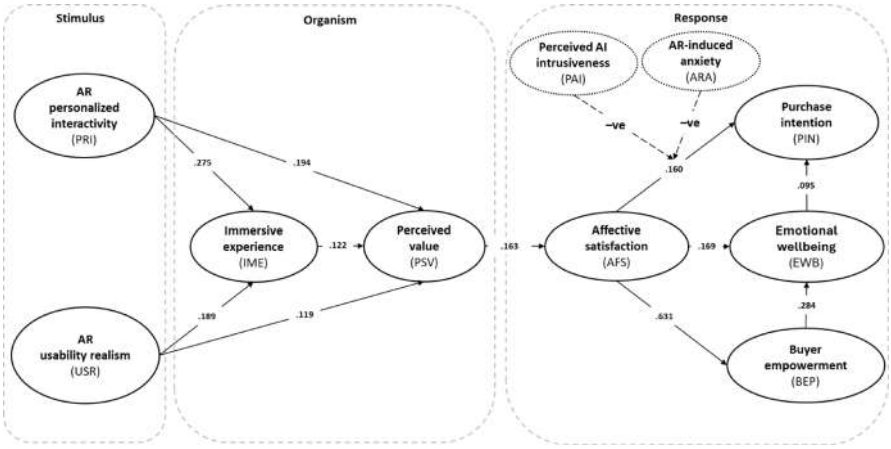


Fig. 4 Structural model

Table 6 Hypothesis testing

Relationship				Standardized estimate	Standard error	Critical ratio	*p*-value	Remarks
H_{1a}	PRI	→	IME	0.275	0.067	4.106	0.000***	Supported
H_{1b}	PRI	→	PSV	0.194	0.054	3.627	0.000***	Supported
H_{2a}	USR	→	IME	0.189	0.077	2.45	0.014*	Supported
H_{2b}	USR	→	PSV	0.119	0.06	1.962	0.05†	Supported
H_3	IME	→	PSV	0.122	0.036	3.344	0.000***	Supported
H_4	PSV	→	AFS	0.163	0.051	3.205	0.001**	Supported
H_5	AFS	→	BEP	0.631	0.059	10.77	0.000***	Supported
H_6	AFS	→	EWB	0.169	0.091	1.868	0.062†	Supported
H_7	AFS	→	PIN	0.16	0.075	2.127	0.033*	Supported
H_8	BEP	→	EWB	0.284	0.075	3.796	0.000***	Supported
H_9	EWB	→	PIN	0.095	0.049	1.946	0.052†	Supported

† = $p<0.10$. * = $p<0.05$. ** = $p<0.01$. *** = $p<0.001$. PRI = AR personalized interactivity. USR = AR usability realism. IME = Immersive experience. PSV = Perceived value. AFS = Affective satisfaction. BEP = Buyer empowerment. EWB = Emotional wellbeing. PIN = Purchase intention

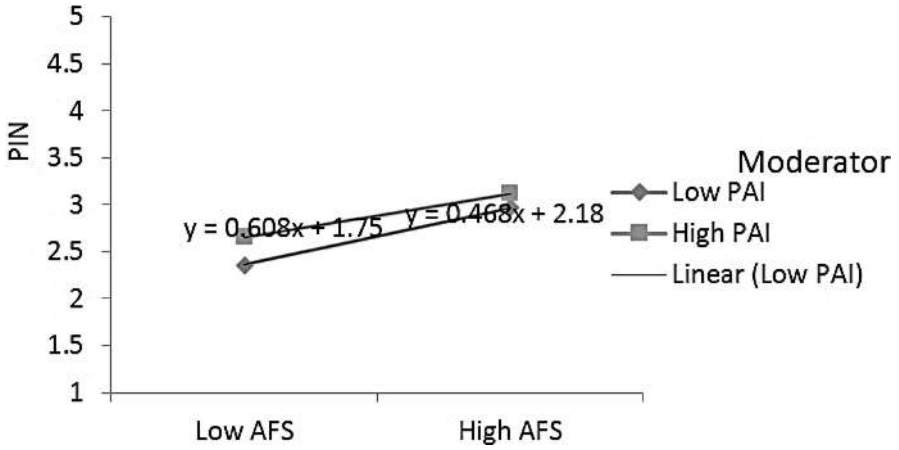


Fig. 5 Moderating effect of perceived AI intrusiveness. Perceived AI intrusiveness (PAI) dampens the positive relationship between affective satisfaction (AFS) and purchase intention (PIN)

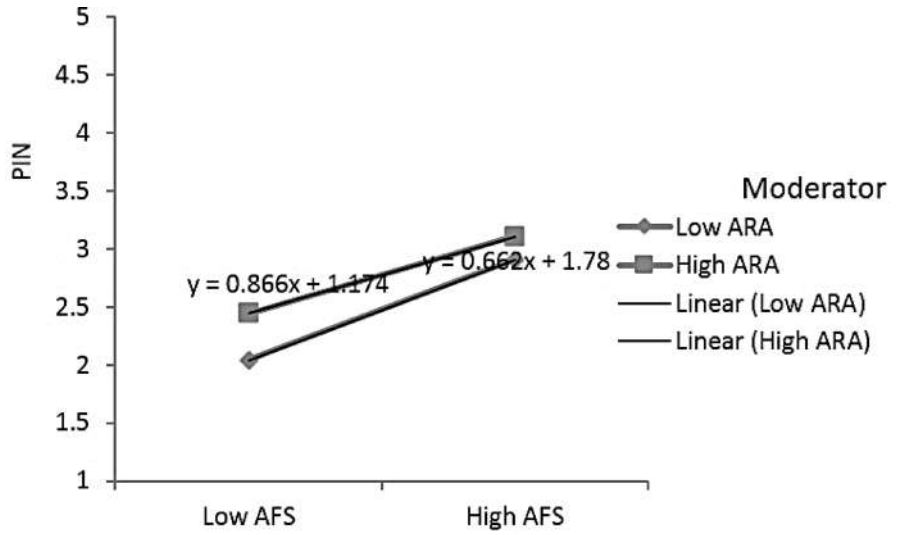


Fig. 6 Moderating effect of AR-induced anxiety. AR-induced anxiety (ARA) dampens the positive relationship between affective satisfaction (AFS) and purchase intention (PIN)

ence of affective satisfaction on purchase intention is reduced under the moderating influence of perceived AI intrusiveness (H_{10a}: $p = 0.05 < 0.10$, Fig. 5) and AR-induced anxiety (H_{10b}: $p = 0.019 < 0.05$, Fig. 6). The next section discusses the implications of these results.

5 Discussion

This research offers empirical support for a consumer-centered model linking AR in online shopping to affective satisfaction, buyer empowerment, emotional wellbeing, and behavioral outcomes. The model integrates insights from prior work, qualitative analysis, and a subsequent quantitative analysis. Adopting a two-phase study design has clarified how users cognitively and emotionally process immersive shopping experiences, particularly regarding interactive personalization and usability-driven realism.

Findings related to H_{1a} and H_{1b} confirm that personalized interactivity through AR fosters both immersive experience and perceived value. These observations build on earlier work highlighting how user control and customized features increase emotional engagement and relevance [51], aligning with S-O-R model [10] and appraisal theory [11]. Similarly, H_{2a} and H_{2b} were supported, indicating that AR usability realism, realized through intuitive interface design and lifelike imagery, generates higher levels of immersion and perceived value. These findings extend previous evidence that seamless navigation and credible visual representations enhance users' sense of presence, utility, and enjoyment [18, 34]. Whereas, H_3 shows that immersion exerts a decisive impact on perceived value, paralleling other studies suggesting that experiential quality influences both hedonic and functional judgments [37], while H_4 underscores that perceived value can elevate affective satisfaction, confirming the role of cognitive appraisals in shaping emotional outcomes [18, 39].

Affective satisfaction was also found to heighten buyer empowerment, emotional wellbeing, and purchase intention (H_5, H_6, and H_7), affirming the view that rewarding interactions with AR can bolster user confidence, positive emotions, and purchasing readiness [35]. H_8 further indicates that buyer empowerment raises emotional wellbeing, linking consumer autonomy and perceived control in technology-enabled contexts [18] while H_9 uncovers a link-albeit modest- between emotional wellbeing and purchase intention, echoing earlier findings regarding the role of psychological safety and positivity in shaping buying decisions [8].

The moderating analyses (H_{10a} and H_{10b}) reveal that perceived AI intrusiveness and AR-induced anxiety diminish the positive effect of affective satisfaction on purchase intention. These results align with concerns about extreme personalization, data intrusion, and excessive sensory demands in immersive technology [8]. Feelings of unease or reduced control can weaken the power of positive emotions to drive behavior, even when satisfaction is high. Cognitive appraisal theory [11] explains why, whereby a positive emotional state raises purchase intention only when the surrounding experience is also judged safe and controllable, which means a secondary appraisal of threat can override the behavioral pull of satisfaction. Perceived AI intrusiveness supplies exactly such a threat appraisal, signaling surveillance or manipulation, whereas AR-induced anxiety signals an inability to cope with sensory or interactional demands. Each appraisal shifts the shopper from an approach orientation toward avoidance even while affective satisfaction remains high. This, in turn, clarifies why emotional gains do not convert into behavior automatically and why psychological safety operates as a gatekeeping condition rather than a peripheral comfort. These insights suggest that designers and managers should consider user comfort and psychological

safeguards when introducing immersive AR features, given the complex interplay among emotional engagement, wellbeing, and perceived intrusiveness. Therefore, this research advances extant understanding of immersive technology adoption by bridging cognitive and emotional factors and emphasizing consumer wellbeing and ethical engagement in the design of AR-enhanced e-commerce.

5.1 Theoretical contributions

Consistent with extant guidelines on theory development [52], this article strengthens theoretical perspectives on digital consumer behavior by bridging the S-O-R model [10] with cognitive appraisal theory [11]. Previous studies often adopted the S-O-R model to account for how environmental cues influence consumer decisions, yet they rarely explained in detail the cognitive filters that shape how individuals process these cues. Hence, by embedding appraisal mechanisms into the organism phase, this article clarifies the pathways through which AR design elements prompt cognitive and emotional responses, including perceived value and affective satisfaction, which then influence buyer empowerment and emotional wellbeing.

The mixed-method approach involving exploratory qualitative and confirmatory quantitative studies not only refines the conceptual understanding of immersive retail, but also highlights the interplay between utilitarian and hedonic appraisals in online shopping. More specifically, this research demonstrates that immersive features—such as personalized interactivity and usability realism—invoke both rational evaluations of functional benefits and emotional assessments related to pleasure, control, and psychological safety [7, 18]. These findings move beyond simpler S-O-R applications by identifying how specific cognitive and emotional triggers can enhance or undermine user wellbeing. The addition of perceived AI intrusiveness and AR-induced anxiety as moderating conditions extends current knowledge on the boundary constraints of personalization [8]. While personalization has often been portrayed as beneficial [53], the findings call attention to adverse psychological effects arising from data intrusiveness or sensory overload. These results connect the study to three wider conversations in concrete terms. The dampening role of perceived AI intrusiveness gives empirical weight to debates on digital ethics and over-personalization, showing that intrusive tailoring is not only normatively questionable but commercially self-defeating once it crosses into discomfort [53]. The positioning of emotional wellbeing as a modeled outcome, rather than an afterthought, situates immersive commerce within transformative service research, which evaluates services by their capacity to support rather than erode consumer wellbeing [15]. Reading the two in combination implies that responsible AR design and commercial performance are complementary rather than opposed, which is the sense in which wellbeing sits at the center of value creation in immersive retail.

This article also speaks to research on consumer wellbeing and ethical usage of immersive technologies by showing that affective reactions and perceptions of control are not mere afterthoughts in AR design [43]. Instead, they are central factors that can strengthen or diminish technology acceptance. Revealing how positive emotions can be offset by intrusiveness or anxiety emphasizes the ethical mandate for responsible innovation. A rich theoretical perspective emerges in which consumer wellbe-

ing serves as both an outcome of immersive design and an influence on purchase intentions. This emphasis on the connections between emotional outcomes, decision confidence, and user acceptance enriches existing discourse on technology-enabled commerce. Hence, by conducting qualitative and quantitative inquiries, this research captures real-world complexity while providing structured evidence of how personalization, immersion, and wellbeing intertwine in AR e-commerce. Consequently, this research deepens understanding of how immersive elements, coupled with ethical considerations, can guide the future development of consumer-focused digital environments.

5.2 Managerial implications

The validated measures (Table 3) and relationships (Table 6) herein offer a set of actionable recommendations for practice. AR developers and managers in e-commerce can leverage these findings by systematically embedding interactivity and realism into their systems. Specifically, offering users clear ways to customize and control the AR environment—such as trying on items or rearranging virtual furniture—prompts stronger engagement and perceived value. The item statements that underscore consumers' desire to tailor AR to their individual preferences (for instance, "I can customize the AR experience according to my preferences") indicate that platforms need easily accessible menus or toggles that adjust the AR settings or level of personalization [31]. Hence, by doing so, brands can encourage a sense of user agency and functional benefit that drives satisfaction and eventual purchase intention.

Realism also emerges as a critical pillar. The study's usability realism indicators show that fluid visuals, intuitive system responses, and lifelike representations boost both immersion and perceived value [32]. This means retail platforms should invest in quality image rendering, minimal lag times, and responsive controls so that users feel the virtual environment is natural and trustworthy. Even small enhancements—such as accurate lighting and scaling—can strengthen consumers' impression that the AR environment closely aligns with their real-world settings, thus promoting deeper engagement and enjoyment.

Affective satisfaction and emotional wellbeing were strongly linked to consumer empowerment and buying decisions. Users who experience positive emotions during AR sessions tend to feel more confident, more in control, and more inclined to purchase [34, 37]. Retailers might foster these beneficial emotions by integrating pleasing visuals, game-like interactions, and immediate feedback mechanisms that affirm user choices and accomplishments. This could include progress indicators, augmented "try-on" confirmations, or interactive messages that highlight how the AR environment is adapting to user preferences.

The findings also reveal potential pitfalls when technology becomes too invasive [46, 47] or induces stress [48]. Excess personalization—where users sense that the system is "learning" too much about them—or confusion arising from overly complex AR interfaces can reduce the likelihood that emotional satisfaction translates into actual purchasing. Providing transparent data-usage guidelines and placing users in charge of the extent and type of personalization can mitigate these risks. For instance,

shoppers may decide whether they want the platform to recommend items based on their past browsing habits or location. Clear communication around data collection and the option to opt out can help maintain a comfortable sense of privacy and trust.

Cultural nuances further complicate AR adoption across international markets [25]. Some shoppers might value detailed customization while others may prefer minimal interference in their browsing. In regions with higher privacy sensitivity, developers should highlight security features, data anonymity, and user autonomy more prominently. For sectors requiring greater realism—like cosmetics, fashion, or furniture—platforms can tailor visuals to reflect local environments, styles, and preferences, allowing users to see items in contexts that feel familiar and personally relevant.

Translating these principles into practice calls for concrete implementation rather than general intent. Platforms can A/B test the intensity of personalization and the default privacy setting against the very outcomes modeled here, comparing immersion, perceived intrusiveness, anxiety, and conversion across an opt-in arm in which personalization is off by default and an opt-out arm in which it is on, which yields direct evidence on where helpful tailoring tips into intrusion. Staged rollouts of new AR features to a small user segment before full release allow the same emotional indicators to be monitored as guardrails, which catches a feature that lifts conversion while raising anxiety before it reaches the whole user base. Visible controls also work best when concrete, such as a one-tap toggle to disable behavioral tracking, a slider that scales recommendation aggressiveness, and a short panel that states what data the session uses, each of which converts the principle of transparent personalization into an interface element a shopper can see and operate.

These considerations, when taken collectively, draw attention to the need for emotional safety and ethical design in retail technology. AR developers who emphasize a balance between personalization and control—rather than one at the expense of the other—are likely to see stronger consumer loyalty and higher purchase intentions. When users perceive that an AR feature supports their decision-making process while respecting their comfort zones, they are more apt to find both short-term and long-term value in digital shopping experiences.

6 Conclusion

This study draws attention to the impact of AR features in shaping affective satisfaction, buyer empowerment, emotional wellbeing, and purchase intention. Personalized, realistic, and intuitive AR features have been found to enhance users' sense of control, foster greater immersion, and strengthen both hedonic and functional appraisals. The S-O-R model [10] integrated with cognitive appraisal theory [11] highlights how individuals' cognitive and emotional responses arise from technologically driven stimuli, which then shape key behavioral outcomes.

Nonetheless, some caution is warranted, given that the cross-sectional nature of this research precludes definitive causal inferences. While international participation was part of the study design, cross-cultural differences were not explicitly tested, so any cultural insights remain speculative. Another limitation involves reliance on self-

selected AR users, which restricts generalizability to more hesitant or inexperienced populations. Future studies might adopt experimental or longitudinal approaches to clarify causal pathways, as well as investigate other emotional consequences, including digital fatigue or shifts in brand trust. Further research on how cultural background moderates AR usage could shed new light on global adoption patterns and user satisfaction. The practical and scholarly need to design immersive technologies that preserve moral alignment and foster psychological comfort remains an urgent priority, especially as digital engagement becomes more prevalent among diverse consumer groups.

The scope of the conclusions also carries product category boundaries that future work should map directly. The four categories examined here (apparel, cosmetics, electronics, furniture) differ in how much they gain from visualization, which implies that the strength of the modeled paths is likely to vary between utilitarian goods that AR helps evaluate through fit and placement and experiential goods whose value rests on sensations that AR cannot fully reproduce [54]. Stating this boundary condition bears on transfer to other settings, since the same immersive feature that reduces uncertainty for a sofa or a pair of glasses may add little for categories defined by taste, scent, or social context. A further boundary concerns measurement, given that the constructs here were captured through validated multi-item self-report scales rather than behavioral or platform-side indicators, an approach standard in this literature yet open to common-method and self-presentation concerns. Purchase intention is itself a stated psychological outcome rather than observed purchase, which keeps the present findings bounded to intention and leaves any behavioral claim contingent on later corroboration [55]. Future studies could supply that corroboration by pairing survey measures with objective behavioral, physiological, or log-based signals of immersion, anxiety, and purchase.

Appendix 1: Interview protocol

The semi-structured protocol below guided the 20 interviews in Study 1. The questions were developed following the guidelines in Lim [26] and were used across all four sites, with minor wording adjustments for local comprehension. Each interview opened with a broad recollection of a recent augmented reality shopping experience and then turned to progressively more specific aspects of that experience, with the order kept flexible so that participants could lead and elaborate in their own terms. Follow-up probes such as “Could you say more about that?” or “What made you feel that way?” were used throughout to surface the cognitive and emotional reflections that the thematic analysis later coded inductively.

1. To begin, could you tell me about a recent time you used an augmented reality feature while shopping online? What were you shopping for, and what did you hope the feature would help you do?
2. Walk me through what the experience was like as you used it. How much could you tailor or adjust it to your own preferences, and how much control did you feel over how it unfolded?

3. How easy or difficult was the experience to navigate? How lifelike did the visuals appear, and did it run smoothly enough to feel natural and trustworthy?
4. While using the feature, how absorbed did you feel? Did you ever lose track of time or feel as though you were inside the scene rather than looking at a screen?
5. How did using it make you feel emotionally? Were there moments of enjoyment, pleasure, or satisfaction, and what brought them on?
6. Were there moments when it made you feel uneasy, nervous, or stressed? If so, what triggered that, and how did you respond?
7. Did any part of it feel intrusive or too personal, for instance, in how much the system seemed to know about you or how it recommended items? Were there moments that felt manipulative or uncomfortable in terms of privacy?
8. Thinking about the time and effort involved, was the experience worth it for you? In what ways, if any, did it add value beyond a standard product page?
9. After using the feature, how did you feel about your purchase decision? Did it make you feel more confident, more in control, or more capable as a shopper?
10. How did the experience affect your willingness to buy? Were you more or less likely to purchase the product afterward, and why?
11. Stepping back from the details, how did the experience leave you feeling overall? Did it leave you feeling more positive and refreshed, or more drained and unsettled? What do you think shaped that?
12. Is there anything about your augmented reality shopping experience, positive or negative, that we have not discussed and that feels important to you? What would make the experience better?

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Data availability Data can be made available upon reasonable request.

Declarations

Conflict of interest Weng Marc Lim is an Associate Editor of *Electronic Commerce Research* and was excluded from editorial decision-making related to the acceptance of this article for publication in the journal.

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